SDN BASED FIREWALL

COMPUTER NETWORKS (ORANGE PROBLEM)

UE24CS252B

NAME : HARSH M

SRN : PES1UG24CS573

CLASS-SECTION : 4-J

CONTEXT:

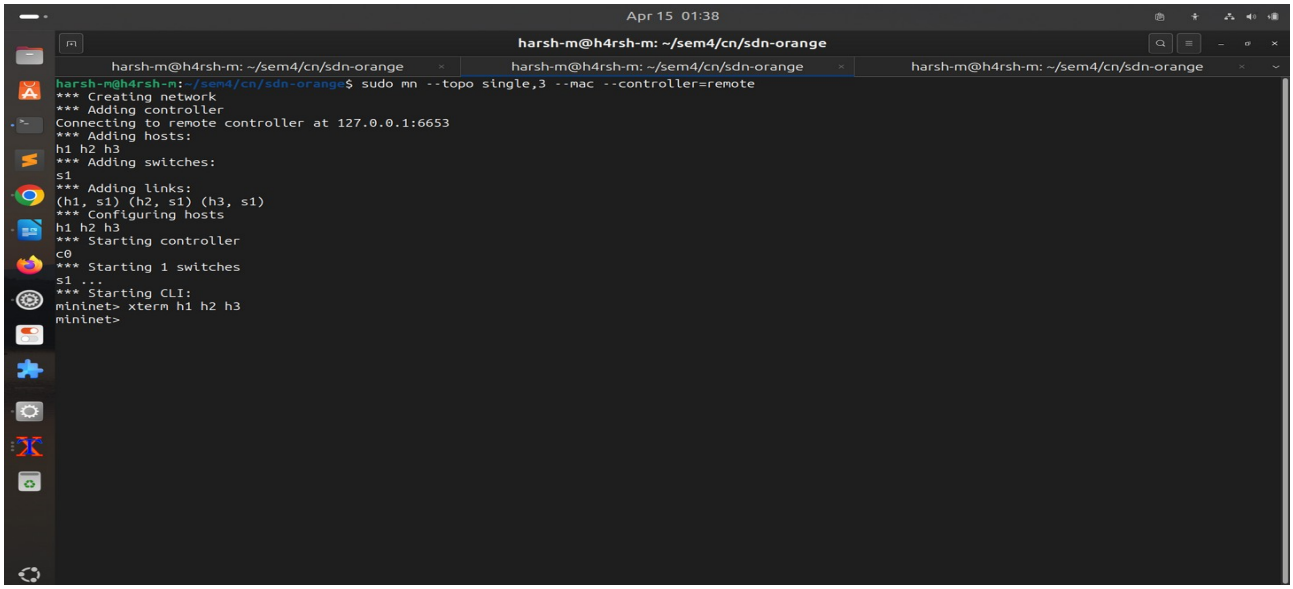
Three terminals are opened :

1. Firewall.py
2. Mininet which has spawned three nodes h1, h2, h3 along with switch s1.
3. Wireshark to track the activity

H1 is treated as malicious , I.e the firewall is supposed to block it.

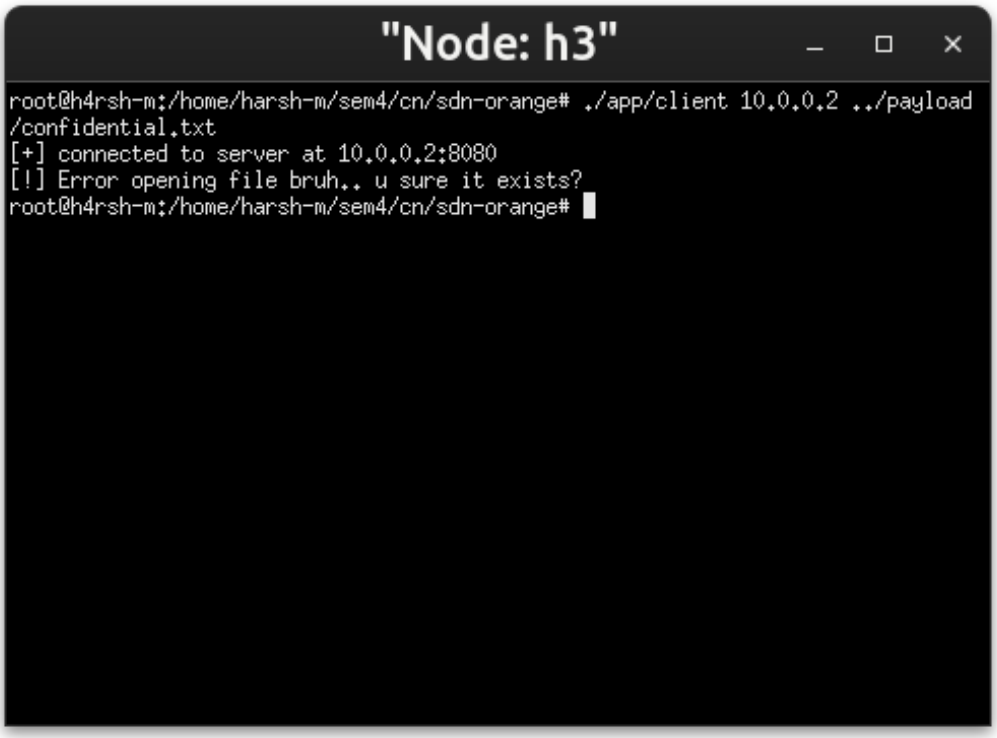
H2 is the receiver, the client, and h3 is the sender that is not blocked.

SCREENSHOTS:



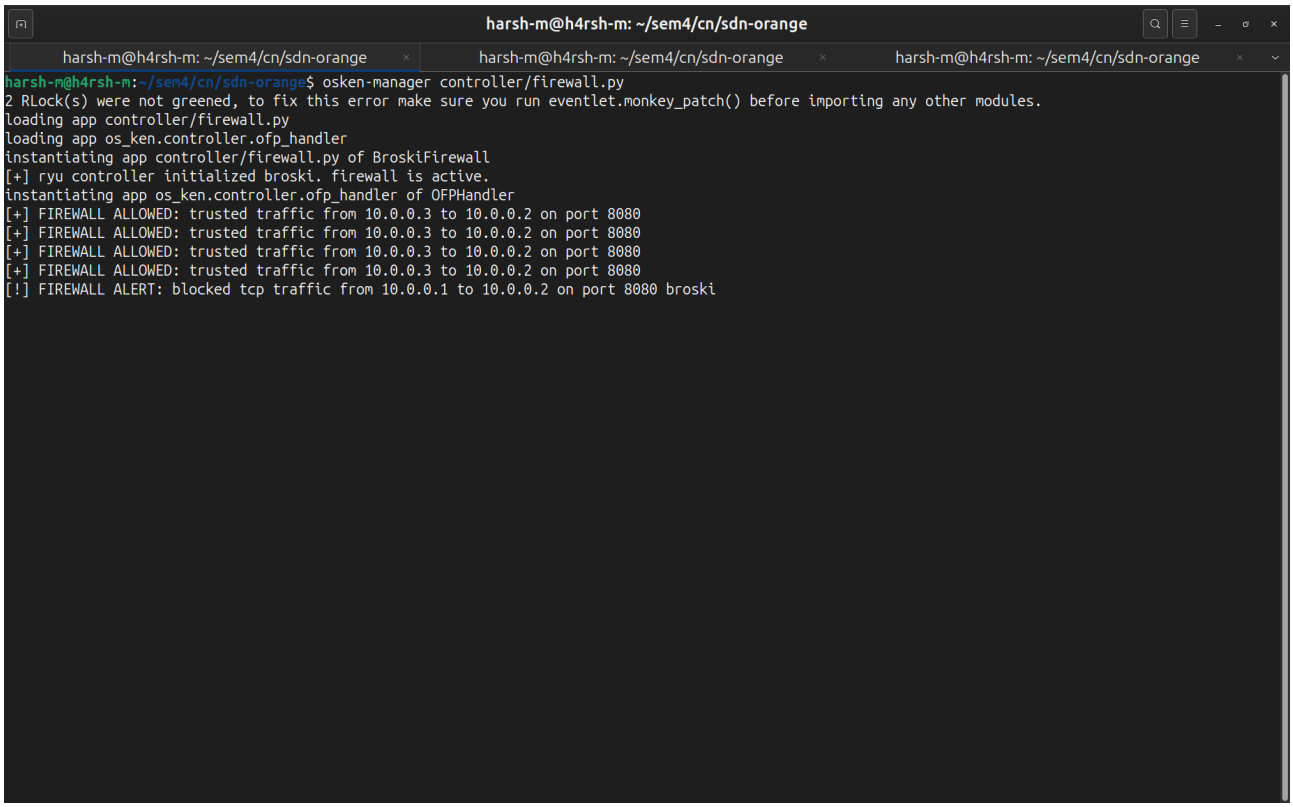
Node : h2:

Node h3 :



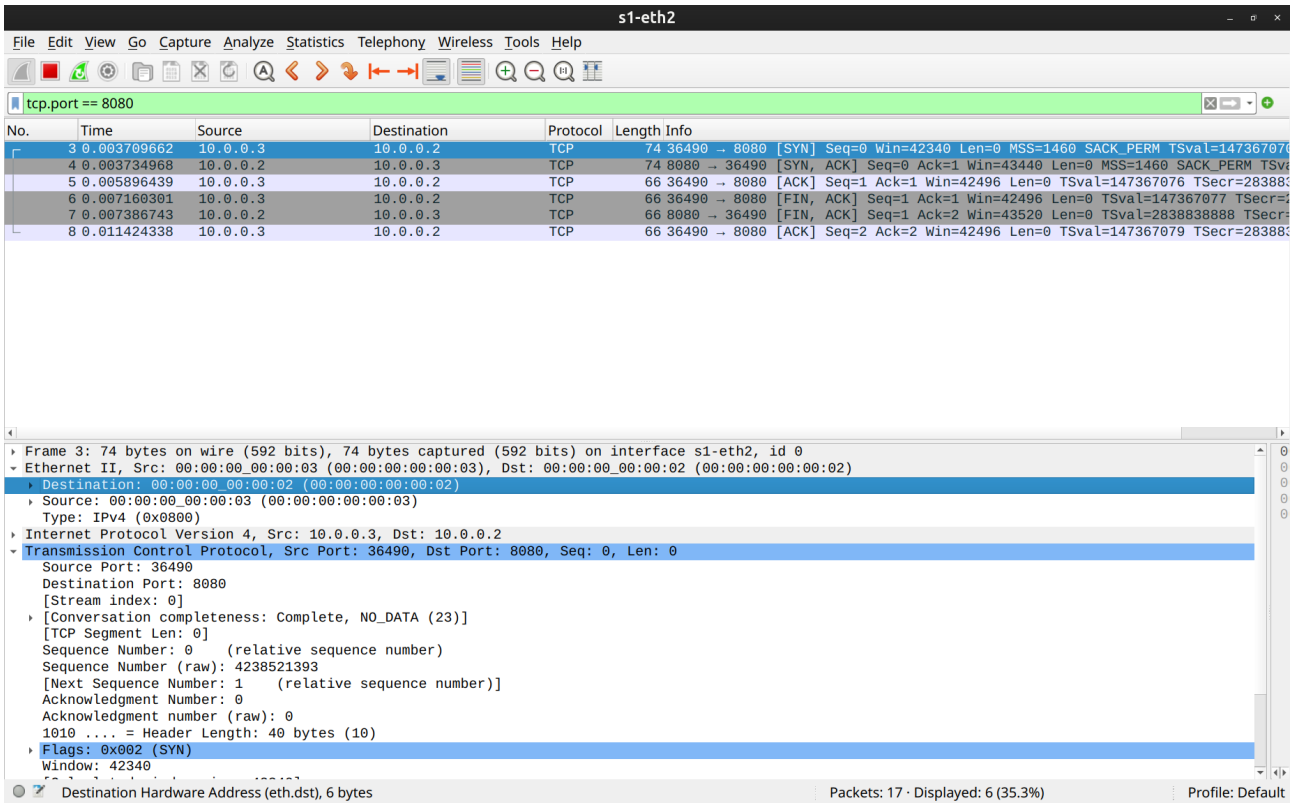
Node h1:

Firewall terminal :



Here we can clearly see allowed vs blocked traffic , from h3 and h1 respectively.

Wireshark capture:



Here we can see the handshake attempt, and responses between h3 (10.0.0.3) and h2 (10.0.0.2) in the form of syn synack ack, and then the text file payload delivery tracked as it is.

What we can't see is the requests from h1(10.0.0.1) since the firewall has blocked them , so they are dropped somewhere into Madhya Pradesh (which nobody cares about).

Ping tests:

```
mininet> pingall
*** Ping: testing ping reachability
h1 -> h2 h3
h2 -> h1 h3
h3 -> h1 h2
*** Results: 0% dropped (6/6 received)
mininet> h1 ping -c 4 h2
PING 10.0.0.2 (10.0.0.2) 56(84) bytes of data.
64 bytes from 10.0.0.2: icmp_seq=1 ttl=64 time=0.253 ms
64 bytes from 10.0.0.2: icmp_seq=2 ttl=64 time=0.121 ms
64 bytes from 10.0.0.2: icmp_seq=3 ttl=64 time=0.212 ms
64 bytes from 10.0.0.2: icmp_seq=4 ttl=64 time=0.089 ms

--- 10.0.0.2 ping statistics ---
4 packets transmitted, 4 received, 0% packet loss, time 3091ms
rtt min/avg/max/mdev = 0.089/0.168/0.253/0.066 ms
mininet> h3 ping -c 4 h2
PING 10.0.0.2 (10.0.0.2) 56(84) bytes of data.
64 bytes from 10.0.0.2: icmp_seq=1 ttl=64 time=0.216 ms
64 bytes from 10.0.0.2: icmp_seq=2 ttl=64 time=0.073 ms
64 bytes from 10.0.0.2: icmp_seq=3 ttl=64 time=0.085 ms
64 bytes from 10.0.0.2: icmp_seq=4 ttl=64 time=0.280 ms

--- 10.0.0.2 ping statistics ---
4 packets transmitted, 4 received, 0% packet loss, time 3146ms
rtt min/avg/max/mdev = 0.073/0.163/0.280/0.087 ms
mininet>
```

Iperf test:

h3 to h2: Result 33.8 Gb/s bandwidth

```
mininet> h2 iperf -s -p 8080 &
mininet> h3 iperf -c 10.0.0.2 -p 8080
------------------------------------------------------------
Client connecting to 10.0.0.2, TCP port 8080
TCP window size: 85.3 KByte (default)
------------------------------------------------------------
[  1] local 10.0.0.3 port 56208 connected with 10.0.0.2 port 8080 (icwnd/mss/irtt=14/1448/252)
[ ID] Interval       Transfer     Bandwidth
[  1] 0.0000-10.0057 sec  39.4 GBytes  33.8 Gbits/sec
mininet>
```

h1 to h2 : Since it is bloacked by the firewall it will hang forever

```
------------------------------------------------------------
[  1] local 10.0.0.3 port 56208 connected with 10.0.0.2 port 8080 (icwnd/mss/irtt=14/1448/252)
[ ID] Interval       Transfer     Bandwidth
[  1] 0.0000-10.0057 sec  39.4 GBytes  33.8 Gbits/sec
mininet> h1 iperf -c 10.0.0.2 -p 8080
------------------------------------------------------------
Client connecting to 10.0.0.2, TCP port 8080
TCP window size: 85.3 KByte (default)
------------------------------------------------------------
```

Flow tables:

```
harsh-m@h4rsh-m:~/sem4/cn/sdn-orange$ sudo ovs-ofctl dump-flows s1
[sudo] password for harsh-m:
 cookie=0x0, duration=3867.895s, table=0, n_packets=31, n_bytes=2294, priority=100,tcp,nw_src=10.0.0.1,nw_dst=10.0.0.2,tp_dst=8080 actions=drop
 cookie=0x0, duration=4259.446s, table=0, n_packets=635014, n_bytes=41913448, priority=1,in_port="s1-eth2",dl_src=00:00:00:00:00:02,dl_dst=00:00:00:00:0
0:03 actions=output:"s1-eth3"
 cookie=0x0, duration=4259.437s, table=0, n_packets=1550587, n_bytes=68811559806, priority=1,in_port="s1-eth3",dl_src=00:00:00:00:00:03,dl_dst=00:00:00:
00:00:02 actions=output:"s1-eth2"
 cookie=0x0, duration=3867.899s, table=0, n_packets=16, n_bytes=1008, priority=1,in_port="s1-eth2",dl_src=00:00:00:00:00:02,dl_dst=00:00:00:00:00:01 act
ions=output:"s1-eth1"
 cookie=0x0, duration=3794.431s, table=0, n_packets=15, n_bytes=966, priority=1,in_port="s1-eth1",dl_src=00:00:00:00:00:01,dl_dst=00:00:00:00:00:02 acti
ons=output:"s1-eth2"
 cookie=0x0, duration=2610.414s, table=0, n_packets=3, n_bytes=238, priority=1,in_port="s1-eth3",dl_src=00:00:00:00:00:03,dl_dst=00:00:00:00:00:01 actio
ns=output:"s1-eth1"
 cookie=0x0, duration=2610.413s, table=0, n_packets=2, n_bytes=140, priority=1,in_port="s1-eth1",dl_src=00:00:00:00:00:01,dl_dst=00:00:00:00:00:03 actio
ns=output:"s1-eth3"
 cookie=0x0, duration=6126.161s, table=0, n_packets=54, n_bytes=3868, priority=0 actions=CONTROLLER:65535
harsh-m@h4rsh-m:~/sem4/cn/sdn-orange$
```